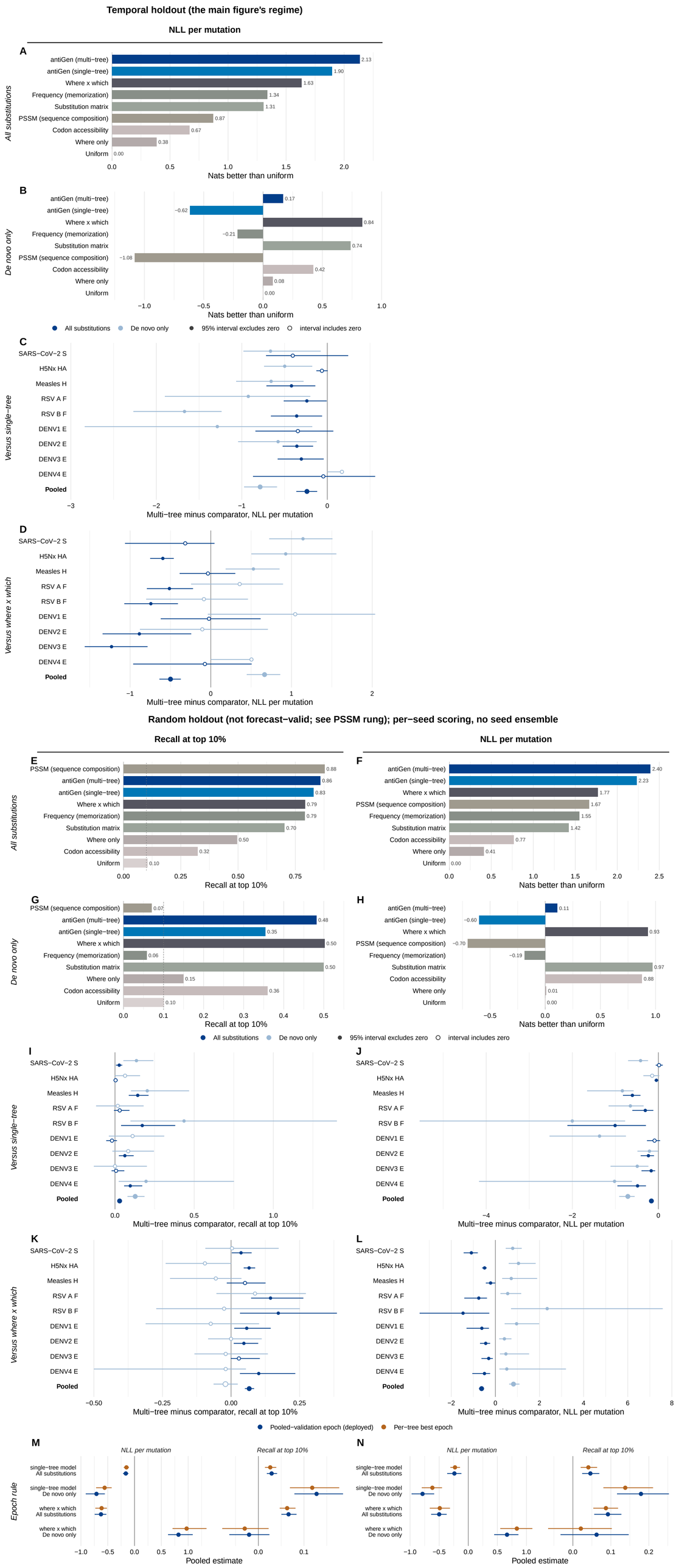
Temporal holdout (the main figure's regime)
NLL per mutation
A
All substitutions
antiGen (multi−tree) 2.13
antiGen (single−tree) 1.90
Where x which 1.63
Frequency (memorization) 1.34
Substitution matrix 1.31
PSSM (sequence composition) 0.87
Codon accessibility 0.67
Where only 0.38
Uniform 0.00
0.0 0.5 1.0 1.5 2.0
Nats better than uniform
B
De novo only
antiGen (multi−tree) 0.17
antiGen (single−tree) −0.62
Where x which 0.84
Frequency (memorization) −0.21
Substitution matrix 0.74
PSSM (sequence composition) −1.08
Codon accessibility 0.42
Where only 0.08
Uniform 0.00
−1.0 −0.5 0.0 0.5 1.0
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
C
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−3 −2 −1 0
Multi−tree minus comparator, NLL per mutation
D
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−1 0 1 2
Multi−tree minus comparator, NLL per mutation
Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble
Recall at top 10%
NLL per mutation
E
All substitutions
PSSM (sequence composition) 0.88
antiGen (multi−tree) 0.86
antiGen (single−tree) 0.83
Where x which 0.79
Frequency (memorization) 0.79
Substitution matrix 0.70
Where only 0.50
Codon accessibility 0.32
Uniform 0.10
0.00 0.25 0.50 0.75
Recall at top 10%
F
antiGen (multi−tree) 2.40
antiGen (single−tree) 2.23
Where x which 1.77
PSSM (sequence composition) 1.67
Frequency (memorization) 1.55
Substitution matrix 1.42
Codon accessibility 0.77
Where only 0.41
Uniform 0.00
0.0 0.5 1.0 1.5 2.0 2.5
Nats better than uniform
G
De novo only
PSSM (sequence composition) 0.07
antiGen (multi−tree) 0.48
antiGen (single−tree) 0.35
Where x which 0.50
Frequency (memorization) 0.06
Substitution matrix 0.50
Where only 0.15
Codon accessibility 0.36
Uniform 0.10
0.0 0.1 0.2 0.3 0.4 0.5
Recall at top 10%
H
antiGen (multi−tree) 0.11
antiGen (single−tree) −0.60
Where x which 0.93
PSSM (sequence composition) −0.70
Frequency (memorization) −0.19
Substitution matrix 0.97
Codon accessibility 0.88
Where only 0.01
Uniform 0.00
−0.5 0.0 0.5 1.0
Nats better than uniform
All substitutions
De novo only
95% interval excludes zero
interval includes zero
I
Versus single−tree
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
0.0 0.5 1.0
Multi−tree minus comparator, recall at top 10%
J
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−4 −2 0
Multi−tree minus comparator, NLL per mutation
K
Versus where x which
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−0.50 −0.25 0.00 0.25
Multi−tree minus comparator, recall at top 10%
L
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
−2 0 2 4 6 8
Multi−tree minus comparator, NLL per mutation
Pooled−validation epoch (deployed)
Per−tree best epoch
M
NLL per mutation
Recall at top 10%
Epoch rule
single−tree model All substitutions
single−tree model De novo only
where x which All substitutions
where x which De novo only
−1.0 −0.5 0.0 0.5 1.0 0.0 0.1
Pooled estimate
N
NLL per mutation
Recall at top 10%
single−tree model All substitutions
single−tree model De novo only
where x which All substitutions
where x which De novo only
−1.0 −0.5 0.0 0.5 1.0 0.0 0.1 0.2
Pooled estimate